

NANTES ATLANTIQUE AP, France

WMO: 072220

Lat: 47.1500N

Lon: 1.6089W

Elev: 27

StdP: 101.00

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.4	-1.9	-7.0	2.1	-1.3	-5.0	2.5	-0.5	12.5	11.2	11.2	11.4	2.3	60	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	30.9	20.1	28.8	19.5	26.8	18.7	21.5	28.3	20.5	26.5	19.6	24.8	3.9	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.1	23.4	18.5	13.4	22.4	17.7	12.7	21.6	62.4	28.4	58.9	26.6	55.8	24.8	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.5	7.5	DB	-5.6	34.8	2.1	1.9	-7.1	36.2	-8.4	37.3	-9.5	38.3	-11.1	39.7	(4)
(5)			WB	-6.2	23.2	2.1	1.0	-7.7	24.0	-9.0	24.6	-10.1	25.2	-11.7	25.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	6.2	6.7	9.0	11.1	14.4	17.7	19.6	19.5	16.8	13.6	9.2	6.6	(6)	
	DBStd	5.83	3.75	3.40	2.99	3.02	3.04	3.24	3.01	2.89	2.84	3.10	3.27	3.89	(7)	
	HDD10.0	473	123	98	55	21	2	0	0	0	0	7	51	116	(8)	
	HDD18.3	2273	376	326	290	217	128	49	20	18	62	149	273	364	(9)	
	CDD10.0	1410	6	6	23	54	138	232	297	293	206	118	28	10	(10)	
	CDD18.3	168	0	0	0	1	6	31	59	53	17	2	0	0	(11)	
	CDH23.3	1496	0	0	0	9	61	297	519	453	147	9	0	0	(12)	
	CDH26.7	461	0	0	0	0	8	89	172	157	35	1	0	0	(13)	
(14) Wind	WSAvg	3.6	4.1	4.0	4.0	3.8	3.6	3.4	3.3	3.1	3.2	3.3	3.5	3.9	(14)	
(15) Precipitation	PrecAvg	828	93	72	56	62	61	45	46	46	62	94	95	96	(15)	
	PrecMax	1098	178	149	151	175	137	124	120	102	220	220	198	170	(16)	
	PrecMin	552	11	5	7	12	13	4	6	9	0	19	25	10	(17)	
	PrecStd	126	47	39	35	38	28	25	24	29	50	43	49	47	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	16.3	20.7	24.6	28.0	32.7	33.8	34.1	30.3	24.4	17.9	14.7	(19)	
		MCWB	12.1	11.6	13.5	16.1	18.4	21.2	20.8	21.3	19.1	18.1	14.8	13.0	(20)	
	2%	DB	13.0	13.8	17.9	21.6	25.1	29.0	30.6	30.5	27.2	21.7	16.2	13.8	(21)	
		MCWB	11.8	10.8	12.3	14.8	17.1	19.8	19.9	20.3	18.3	17.1	14.2	12.6	(22)	
	5%	DB	12.2	12.4	15.7	18.9	22.9	26.7	28.5	28.1	24.7	19.7	15.1	13.0	(23)	
		MCWB	11.1	10.4	11.4	13.1	16.5	18.8	19.2	19.5	17.5	16.2	13.5	11.9	(24)	
	10%	DB	11.4	11.5	13.8	16.7	20.5	24.4	26.2	25.6	22.5	18.2	14.1	12.2	(25)	
		MCWB	10.3	10.0	10.6	12.1	15.2	17.7	18.4	18.5	16.4	15.6	12.5	11.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	12.5	14.3	17.0	20.0	22.7	22.4	23.3	20.8	19.0	16.1	13.7	(27)	
		MCDB	13.2	14.9	18.8	23.0	25.9	29.7	30.2	31.1	26.1	21.7	16.8	14.1	(28)	
	2%	WB	12.1	11.6	13.0	15.2	18.2	20.9	21.0	21.6	19.5	17.8	14.7	12.9	(29)	
		MCDB	12.7	12.9	16.3	20.2	23.4	27.4	28.3	28.1	24.6	20.3	15.7	13.5	(30)	
	5%	WB	11.4	11.0	12.1	13.9	16.9	19.5	20.1	20.5	18.5	17.0	13.8	12.1	(31)	
		MCDB	12.1	12.0	14.5	17.8	21.7	25.3	26.3	25.9	22.9	19.1	14.9	12.9	(32)	
	10%	WB	10.5	10.3	11.3	12.6	15.7	18.3	19.2	19.4	17.5	16.1	12.8	11.3	(33)	
		MCDB	11.3	11.4	13.3	15.8	19.7	23.3	24.6	23.7	21.2	17.8	14.0	12.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	7.1	8.4	9.3	9.4	10.1	10.4	10.7	10.4	8.2	6.8	6.0	(35)	
		MCDBR	5.4	7.9	11.1	12.5	12.8	13.6	13.9	14.5	13.6	9.5	7.0	5.6	(36)	
	5% WB	MCWBR	4.6	5.8	6.8	6.8	6.7	6.2	5.6	5.9	6.1	5.6	5.2	4.8	(37)	
		MCWBR	5.1	6.8	8.8	11.1	11.6	12.4	12.1	12.3	11.4	8.1	6.2	5.5	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.330	0.346	0.374	0.393	0.397	0.404	0.397	0.386	0.367	0.364	0.339	0.324	(40)	
		taud	2.427	2.384	2.304	2.254	2.291	2.307	2.328	2.376	2.408	2.419	2.429	2.406	(41)	
	Ebn at Noon	Ebn at Noon	749	814	842	859	868	861	862	858	842	782	730	706	(42)	
		Edh at Noon	69	88	110	127	127	126	122	111	98	84	69	63	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.11	2.02	3.16	4.61	5.45	6.07	5.80	5.14	4.06	2.37	1.37	0.97	(44)	
	RadStd	RadStd	0.10	0.22	0.35	0.49	0.41	0.54	0.31	0.41	0.29	0.19	0.13	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page